Presenting the Cloud Resource Lab

A step-by-step guide to using the simulation alongside the 15-slide deck *Cloud Resource Management, Scheduling, and Cost Optimization*, plus a review of what is missing or worth improving.

The short version. The simulation covers slides 4, 5, 7, 10, 12, 14 and 15 well. Slides 2, 3, 6, 8, 9 and 11 are only partly covered or covered by text on the Learn tab, and the app cannot show vertical scaling, throughput, SLA or availability. Slide 14 still says the demo link will be added after deploy, and the GitHub Pages address I tried returned **404**, so the app does not look published yet. Section 6 lists the fixes in priority order.

1. Before you present (checklist)

- **Deploy it.** Push to `main`; the workflow in `.github/workflows/deploy.yml` publishes to GitHub Pages (Settings > Pages > Source: GitHub Actions). Then put the real URL (and a QR code) on slide 14.
- **Keep a local backup.** Run `npm run dev` in the project folder and keep that tab open in case the venue Wi-Fi fails.
- **Turn on Large text** (A+ button, top right) for the projector. The browser remembers it.
- **Start on the Guided demo tab** and leave the seed alone. Seed 16 is the one whose numbers are quoted below.
- **Rehearse once with the numbers in this guide** so you recognise them on screen. Every figure comes from running the engine itself (spike workload, seed 16, 1 to 4 servers). The 36 unit tests pass.

2. Suggested run of show (15 minutes)

Slides	Min	On screen	What you do
1-3 Title, lifecycle, goals	2	Slides only	Frame the four goals: availability, low cost, fairness, performance. Say the lab lets them *see* cost, wait and utilization trade against each other.
4-6 Provisioning, auto-scaling, metrics	2.5	Simulate tab	Play the default run. Point at servers, queue and the three charts. Tie it to the 70 / 30 / cooldown example.
7-9 Scheduling	2	Simulate tab, Scheduler example	Show the A=10, B=2, C=1 table: FCFS 7.3 min vs SJF 1.3 min. Cover scheduling levels and advanced methods by speaking.
10-12 Pricing, cost, trade-offs	2.5	Learn tab (pricing table), Trade-off calculator	Show the AWS / Azure / Google name table, then the $9.22 / $36.86 / $11.52 calculator.
13 Research	0.5	Slides only	Name the three papers and one takeaway each.
14 Simulation demo	4	Guided demo, steps 1-4	Walk the four steps (section 4). This is the centre of the talk.
15 Discussion	1.5	Discussion tab	Ask both questions, take answers, then press Show guidance.

If you run short, trim slides 8-9 first and keep the demo whole.

3. Step by step, topic by topic

Part A - Slides 2 to 6: resources, provisioning, auto-scaling, metrics (Simulate tab)

- Open **Simulate**. It loads the default setup: spike workload, 1 starting server, up to 4, auto-scaling on, FCFS, on-demand. That is exactly Step 2 of the guided demo.
- Press **Play** at speed 2 or 5. Say: *"Each box is a server. Tasks wait in the queue. One tick is one minute."*
- Slide 4 (static vs dynamic): turn **Auto-scaling off** and capacity stays at 1 server. That is static provisioning. Turn it back on for dynamic provisioning.
- Slide 5 (the 70% / 30% example): open the auto-scaling controls. The defaults already match the slide: scale out above 70, scale in below 30, cooldown 5 min. The events panel shows lines such as *"Requested S2. Ready in 2 min."* then *"S2 is ready."* That 2-minute gap is the delay slide 12 mentions.
- Slide 6 (metrics): in Results, **Average wait** and **Average completion** stand in for latency, and **Server utilization** is utilization (the CPU utilization chart is the live version). Throughput and SLA are **not** shown (see issue 4). Say them aloud with an example.

Part B - Slides 7 to 9: scheduling (Simulate tab)

- Scroll to **Scheduler example**. Three tasks arrive together: A=10, B=2, C=1 minutes. FCFS (A, B, C) gives waits 0, 10, 12 = **7.3 min** average. SJF (C, B, A) gives 0, 1, 3 = **1.3 min**. Say: *"Same work, same server, different order."*
- Switch the scheduler in the controls to show all four on the live run. With the default setup the average waits are **FCFS 3.9 min, SJF 3.0, Round Robin 2.3, Priority 4.1**. Round Robin has the lowest wait but a higher completion time (9.0 vs 7.7 min), because tasks get interrupted.
- Name the downside of each: SJF can starve long tasks, FCFS lets a long task block short ones, Priority needs a business reason.
- Slides 8 and 9 (scheduling levels; heuristic, machine learning, metaheuristic): the app has one level only (tasks onto servers). Cover these by speaking or the Learn cards, and say plainly that the lab does not run genetic algorithms or learning-based schedulers.

Part C - Slides 10 to 12: pricing, cost methods, trade-offs

- Open **Learn** and show the pricing table (pay-as-you-go, commitment, spare capacity for AWS, Azure, Google Cloud). It matches slide 10.
- Back on Simulate, scroll to the **Trade-off calculator**. Rate $0.384 per server hour: fixed 1 server, 24 h = **$9.22**; fixed 4 servers, 96 h = **$36.86** (+300%); auto-scale 1 to 4, 30 h = **$11.52** (+25%). Edit the hours live to show the answer move.
- Say clearly that the calculator is a *daily* estimate. The simulation runs about 100-215 simulated minutes, so its costs are $0.63 to $2.12 at the same rate.
- Slide 11 (cost methods): the lab has right-sizing (starting and max servers), auto-scaling, spot, and shutting down idle servers (scale-in). Storage tiers and FinOps are not modelled; say them aloud.

4. The guided demo (slide 14), four steps

Open the **Guided demo** tab. The workload and seed are identical in every step, so each change gets the credit or blame. Click **Next step**, read the highlighted sentence aloud, then point at the five numbers.

Step	Setup	Avg wait	Cost	Util.	Restarts	Peak servers
1 Auto-scaling off	1 server, FCFS, on-demand	61.1 min	$1.38	100%	0	1
2 Auto-scaling on	1 to 4 servers	3.9 min	$2.12	67%	0	4
3 Switch to spot	same, spot rate	3.9 min	$0.63	67%	1	4
4 Round Robin	same, 2-min slice	2.3 min	$0.63	68%	1	4

Step 1 - Auto-scaling off. The queue peaks at 33 tasks and the last task finishes at minute 215. *"One server is cheap, $1.38, and fully busy, but people wait an hour."* This is slide 12's 'Overloaded at peak'.

Step 2 - Auto-scaling on. Average wait drops **94%** (61.1 to 3.9 min) and the run ends at minute 101. Cost **rises 54%** to $2.12 because more servers run. This matches the slide: wait falls, cost rises. Mention the 2-minute start delay.

Step 3 - Switch to spot. Cost falls **70%** to $0.63 with the same wait, but task T37 is interrupted at minute 60 and restarts from zero. *"Cheaper, but there is a risk."* This also answers discussion question 2.

Step 4 - Round Robin. The longest wait drops from 17 to 8 min and the average to 2.3 min, because every task gets a turn. Add the caveat that average *completion* rises (7.8 to 9.1 min). Round Robin is fair, not faster overall.

Optional extras if time allows

- Press **Open this setup in the simulator** on Step 2, then **Play** to show servers appearing and disappearing in the live view.
- **Compare tab**, slide 12 live: Setup A = fixed 4 servers, auto-scaling off; Setup B = 1 start / 4 max, auto-scaling on. A: wait 3.3 min, cost $2.59, utilization 53%. B: wait 3.9 min, cost $2.12, utilization 67%. *"Slightly slower, 18% cheaper, far less idle capacity."*
- Try **Reserved** pricing on the default run: $1.88 against $2.12 on-demand, because only the starting server gets the lower rate.

If a spot run shows 0 restarts. The interruption chance is 5% per server-hour, checked once per hour, so most seeds show none (in my test of 8 batch seeds, 6 showed none). Keep seed 16 on the demo tab. If you change the seed on the Simulate tab, raise *Spot interruption chance* to 25-50% to make the risk visible.

5. Does the simulation cover the deck?

Slide	Topic	In the simulation?	Notes
2	Resource types and lifecycle	Partial	Only compute (servers). Memory, network, storage are text only. Lifecycle wording differs from the slide (issue 2).
3	Goals: availability, low cost, fairness, performance	Partial	Cost and performance are measured. Fairness shows only through max wait. Availability is not modelled.
4	Static vs dynamic provisioning	Covered	Auto-scaling off / on, but the screen never says 'static' or 'dynamic' (issue 6).
5	Horizontal vs vertical, 70/30 policy	Partial	Horizontal and the 70/30 + cooldown policy are exact. Vertical scaling is not possible: each server has one task slot.
6	Latency, throughput, utilization, SLA	Partial	Wait, completion and utilization shown. No throughput, no SLA.
7	FCFS, Round Robin, Priority, SJF	Covered	All four, plus the 7.3 vs 1.3 example.
8	Scheduling at several levels	Text only	Learn card. The engine has a single level.
9	Heuristic, ML, metaheuristic	Text only	Learn card talks about deadlines and backfilling instead of the slide's three families (issue 3).
10	Pricing models and provider names	Covered	On-demand, reserved and spot rates plus the provider table. The free tier is not mentioned.
11	Cost optimization methods	Partial	Rightsizing, auto-scaling, spot and idle shutdown covered. Storage tiers and FinOps absent.
12	Trade-off numbers	Covered	Calculator reproduces $9.22 / $36.86 / $11.52 exactly.
13	Recent research	Not needed	Slide-only content.
14	Demo bullets	Covered	All four bullets are reproduced by the four demo steps. The live link is not published yet.
15	Discussion questions	Covered	Both questions with hidden guidance.

6. What to fix or improve, in priority order

A. Fix before you present (small effort, visible to the class)

#	Issue	Why it matters	Suggested fix
1	App is not published. Slide 14 says 'link added after deploy'; the Pages URL I tried returned 404.	No live demo link for the class.	Push, enable Pages with GitHub Actions, test the URL on a phone, add the link and a QR code to slide 14.
2	Lifecycle mismatch. Slide 2: Request, Provision, Use, Monitor, Release. Learn tab and glossary: create, use, stop, delete.	The class sees two definitions.	Change the Learn card and the 'Lifecycle' term in `terms.ts` to the five stages.
3	Advanced scheduling mismatch. Slide 9 lists heuristic, machine learning and metaheuristic (genetic algorithm, PSO). The Learn card talks about deadlines and backfilling.	Slide content is not reflected anywhere in the app.	Rewrite the Learn card around the three families; add a line on the research slide's four groups.
4	Throughput and SLA are missing from Results although slide 6 defines both.	A slide topic with nothing to point at.	Add 'Throughput (tasks per hour)' and 'SLA met: % of tasks that waited 5 min or less'. Both can be computed from the existing task list.
5	Cost scale. Slide 12 is dollars per day ($9-$37); the simulator shows $0.63-$2.12.	The audience may think the numbers disagree.	Add a 'scaled to 24 hours' line under Total cost, or state the run length beside it.
6	Static / dynamic are not labelled and vertical scaling is not possible.	Slides 4 and 5 name both.	Label the toggle 'Dynamic (auto-scaling) / Static'. Add a server-size control as a simple vertical-scaling demo, or state on screen that vertical is not simulated.

B. Behaviour of the engine (improves accuracy)

#	Observation	Evidence	Suggestion
7	Scaling thrash. With the default 5-minute cooldown the scaler adds a server and removes it a few minutes later (S4 requested at minute 19, removed at 24; S5 at 29, removed at 34).	Seed 16 spike: cooldown 0 = 8 out / 7 in actions; 5 = 5 / 4; 10 = 3 / 2. Cooldown 10 also gave the lowest wait (3.3 min) at the same cost.	Default the cooldown to 10 min and tell the story: 'cooldown stops this flapping'. Optionally average utilization over 3 ticks.
8	Reserved pricing has no commitment risk. A reservation is paid for the whole term even if idle, but the model charges only for simulated minutes.	Reserved $1.88 vs $2.12 on-demand; the saving is always positive.	Add a note, or bill reserved servers for a minimum period, so over-committing can be shown.
9	Spot interruption is rare and gentle. One 5% chance per server-hour; the server stays and the task simply restarts. No checkpoint option, although discussion question 2 recommends checkpoints.	6 of 8 batch seeds had zero restarts.	Add a 'checkpoint' switch (restart from the last saved minute) and a replacement delay after an interruption.
10	Reserved is not in the guided demo, yet discussion question 1 recommends commitment for the base load.	Demo steps use on-demand and spot only.	Add an optional step 5: 'Reserve the base server'.
11	The step 4 sentence mentions only the longest wait improving.	Average completion rises from 7.8 to 9.1 min.	Add 'but average completion rises' to the sentence so the trade-off is honest.
12	Terminology: the control says 'Scale-out delay', the glossary says 'Start delay'.	SettingsPanel vs terms.ts.	Use one name.

C. Nice to have

- A **Learn card on storage tiers and FinOps**; both are on slide 11 and absent from the app.

- A **presenter mode** (keyboard shortcuts for Next / Back / Play) so you do not need the mouse on stage.
- A **QR code** in the footer pointing to the deployed site.
- Memory and network as extra server properties, only if you want the lab to cover slide 2 beyond compute.

What I checked. I read the deck PDF, the engine, pages and tests, ran the engine for every number quoted here, and ran the unit tests (36 pass). I did not click through the interface in a browser and I did not change any simulation code.